

Load Balancing — Guided Notes

Unit 3.2.2 | CompTIA Network+ N10-009 Obj. 3.2 | N10-008 3.2

Name: _____ Date: _____ Period: _____

Complete each question using the lesson reading. Write in the blank or answer in the space provided.

Question 1

A load balancer sits between _____ and a pool of _____. Clients connect to the load balancer's _____ (VIP) address and never communicate directly with individual backend servers.

Question 2

Layer 4 load balancing makes forwarding decisions based on _____ and _____ — it does not inspect application content. Layer 7 load balancing can inspect _____, URLs, and cookies, which allows routing decisions like 'send /api/ requests to one server pool and /static/ to another.'

Question 3

Match each scheduling algorithm to its description:

Round robin — _____

Least connections — _____

IP hash — _____

Weighted round robin — _____

Question 4

A Layer 7 health check sends an actual _____ request (e.g., GET /health) and checks the _____ code. Why is this more reliable than a Layer 3/4 TCP health check?

Question 5

Sticky sessions (session persistence) ensure requests from the same client always reach the _____ backend server. Why are sticky sessions a workaround rather than an ideal solution?

Question 6

DNS round robin assigns multiple _____ records to the same hostname. List two significant limitations that make it unsuitable as the sole availability mechanism.

Question 7

Global Server Load Balancing (GSLB) distributes traffic across multiple _____ or _____. It typically works through _____ — returning different IP addresses based on client location or data center health.

Question 8 — Real World Application

REAL WORLD: A school's web portal is slow during exam week when all students log in simultaneously. The IT team wants to add load balancing. They have three identical servers available. Describe: which load balancing algorithm you would recommend, whether sticky sessions are needed, and what type of health check you would configure.

Download the answer key from the class server:

```
wget http://[SERVER]/resources/network-engineer/3-2-2-load-balancing/3-2-2-load-balancing-answer-key.pdf
```